THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Who first observed and named ‘cells’ in a thin slice of cork? 1

(a) Schleiden **(b)** Virchow **(c)** Schwann **(d)** Robert Hooke

2. Organisms made up of a single cell are called: 1

(a) multicellular **(b)** colonial only **(c)** tissue-based **(d)** unicellular

3. The cell membrane is described as selectively permeable because it: 1

(a) is made of cellulose **(b)** allows nothing

(c) allows only some substances to pass **(d)** allows all substances

4. Diffusion is the movement of particles from: 1

(a) lower to higher concentration **(b)** higher to lower concentration

(c) outside to inside only **(d)** equal to equal concentration

5. In a concentrated sugar solution a plant cell's cytoplasm shrinks away from the wall. This is: 1

(a) mitosis **(b)** osmosis in reverse

(c) diffusion of cell wall **(d)** plasmolysis

6. The genetic material in a bacterial cell lies in the: 1

(a) plastid **(b)** nucleoid **(c)** vacuole **(d)** nucleolus

7. The energy currency of the cell is: 1

(a) glucose only **(b)** ATP **(c)** RNA **(d)** DNA

8. Plastids that contain chlorophyll are called: 1

(a) chromoplasts **(b)** leucoplasts **(c)** amyloplasts **(d)** chloroplasts

9. The large central vacuole of a plant cell contains: 1

(a) chlorophyll **(b)** ATP **(c)** DNA **(d)** cell sap

10. The ability of a plant cell to grow into a complete plant is called: 1

(a) plasmolysis **(b)** osmosis **(c)** totipotency **(d)** mitosis

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the cell theory in brief. 2

12. Who discovered cells? What did he observe? 2

13. Differentiate between unicellular and multicellular organisms with examples. 2

14. Calculate the total magnification when the eyepiece is 10X and the objective is 40X. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe how the cell membrane is a fluid mosaic. 3

16. Describe an activity to demonstrate osmosis using potato pieces. 3

17. A raisin is placed in water and another in a strong sugar solution. Predict and explain. 3

OR

State three differences between plant and animal cells. 3

18. What is plasmolysis? Describe how it can be observed in a Rhoeo leaf peel. 3

OR

Why do plant cells need a cell wall while animal cells do not? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State the cell theory. 5
(ii) Who contributed to it?
(iii) Why is the cell called the basic unit of life?

OR

(i) How does a light microscope work? 5
(ii) Define resolution and contrast.
(iii) Find the magnification with 10X eyepiece and 100X objective.

20. **Case study:** Potato pieces of equal weight are placed in water and in 20% salt solution for two hours. 5
(i) What happens to each?
(ii) Explain with osmosis.
(iii) Why were the pieces of equal size?
(iv) What would happen in an isotonic solution?

OR

(i) Describe the structure of the cell membrane. 5
(ii) How does it control the passage of materials?
(iii) Distinguish diffusion and osmosis.

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND